2022 / Exam:

Q1:

This question is worth 3% of the course.

Consider the case of a vehicle whose pose (x,y, heading) we want to estimate. Suppose we have a sensor that measures the platform heading (e.g., a compass), whose measurements we want to exploit to implement an EKF update.

The measurements are polluted by WGN of standard deviation = **0.6 degrees**.

State the **H** and **R** matrixes which you would use in the implementation of the EKF update.

You are required to write the derivations of the matrixes on paper.
In your calculations, assume that the estimates of the pose are expressed in (meters; meters; degrees)

The marks allocated for this question are:

- 1 mark for correct H matrix
- 1 mark for correct R matrix
- 1 mark for explanation and derivation

Your solution must be provided in a file named "Q1.pdf" or "Q1.jpg". In the first lines of your file, you must write your name and ZID.

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Q2:

This question is worth 6 marks of the course.

Given the following state space model,

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A} \cdot \mathbf{x}(t) + \mathbf{B} \cdot \mathbf{u}(t)$$

in which matrixes A and B are:

$$A = \begin{bmatrix} -1.2 & -0.2 & -0.2 \\ -0.2 & -0.8 & -0.2 \\ -0.2 & -0.2 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$$

1. Obtain an approximate discrete-time model, for a sampling period = **2** milliseconds.
2. Assuming that we accurately know that its initial state, X(0), has all its elements equal to **2**, and that we also know that the input sequence, **u(k)**, is polluted by WGN of std = **0.1,** you are required to write a program for predicting the state X at discrete time **k=23**. The prediction must be given via an **expected value** and a **covariance matrix**. Your program must be named **"Q2.m"**.
   Assume that your program will be called having the following convention: **[X,P]=Q2(u)**, in which the input variable **u** is an array which contains the input sequence, and **X** and **P** are your result. **No other input parameters are expected. You must respect this specified convention.**

   We recommend writing explanations in the program comments, and submit the program. You also need to write the explanation and derivation on paper, and submit it. Raise any assumptions you made.

   Your explanations must be provided in a file named "Q2.pdf" or "Q2.jpg". In the first lines of your file, you must write your name and ZID. Program file Q2.m must contain a line indicating your name and ZID, as well.

The marks allocation for this question are:

- 2 marks for question 1
- 4 marks for question 2

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Q3:

This question is worth 10% of the course.

Consider a plant whose approximate dynamic model, in continuous time, is given by the following ODE and output equation models

$$\ddot{z}(t) = -4 * z(t) - b * \dot{z}(t) + 16 * u(t)$$
$$y(t) = -4 * z(t) + 2 * \dot{z}(t)$$

in which **u(t)** and **y(t)** are the input and the output of the system, respectively.

The parameter **b** is unknown, and it must be estimated.

An experiment was performed in which the plant was exposed to a given input **u(t)**. Samples of the input signal, **u(t)**, and from the plant's output, **y(t)**, have been collected at a sampling period of 4 ms. We assume that the applied sampling period is small enough for defining an adequate discrete time model by applying the Euler approximation.

You are required to propose a parameter estimation procedure, to be based on exploiting the available measurements, and on using an optimizer (such as "fminsearch", or similar one in Matlab).

Assuming you have been provided a Matlab struct named "Data", having the following fields:

- Data.u : samples of input signal
- Data.y : samples of plant's output
- Data.n : number of samples
- Data.Z0 : the initial value of z
- Data.ZDot0 : the initial value of $\dot{z}$

In which Data.u(k) is the measured input signal at discrete time k, and Data.y(k) the measurement of the plant's output at the same time.

You are required to write a program, which would be used for performing the parameter estimation process. When writing your proposed solution, do not be too concerned about minor syntax errors, we will focus on the logic of the program. However, you must be clear about the equations you are applying in the implementation (e.g. equations and models being used)

We recommend writing those models and explanations in comments in your program. You also need to write the explanation and derivation on paper and submit it. Raise any assumptions you made. Your program must be named "Q3.m". In the first line of it, you must write your name and your ZID.

Assume your program is implemented to run as a function, as follows

"**b=Q3(data);**"

in which the input argument is the data to be used, and the returned value **b** is the estimated parameter.

The mark allocation for this question is:

- 6 marks on the correctness of the cost function
- 4 marks on explanation

If you do not know how to obtain a proper discrete time state space model for solving this question, assume it to be the following one:

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.9 & b \\ 0 & 0.9 \end{bmatrix} \cdot \mathbf{x}(k) + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot u(k)$$

$$y(k) = \begin{bmatrix} 5 & 0 \end{bmatrix} \cdot \mathbf{x}(k)$$

If you used this model, your mark in this question would be reduced by 20%.

Your explanations must be provided in a file named "Q3.pdf" or "Q3.jpg". In the first lines of your file, you must write your name and ZID.

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Q4:

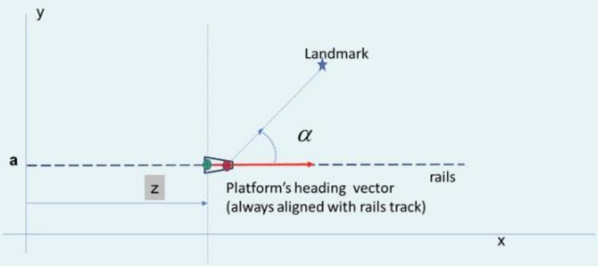
This question is worth 13% of the course.

A locomotive is operated in a 200 meters linear segment of rails, as shown in the figure.

The rail is located parallel to the x-axis (of the GCF), at y=**a=4** meters

For helping in estimating its position, ***z***, the vehicle has an onboard LiDAR which is installed perfectly aligned with the vehicle's heading, but shifted a distance Lx=**0.3**meters (shown as a red dot)

The LiDAR is used to detect landmarks which have been deployed in the area.



(The position of landmark on the image may not be matching with the one stated in the question)

The locomotive's 1D dynamics is approximated by the following model:

$$\ddot{z}(t) = -2 * \dot{z}(t) + 0.4 * u(t)$$

In which **z(t)** is the 1D position of the locomotive, and **u(t)** the voltage being applied to the power amplifier which drives the electric motor which gives traction to the vehicle.

We have interest in estimating the vehicle's 1D position, ***z***, by applying the EKF approach.

The amplitude of the input voltage applied is known from a sensor which introduces an error which is assumed to be WGN, of standard deviation equal to **0.2** volts.

The position, z, is expressed in meters. Assume that all the parameters in the model are consistent with the engineering units being used for ***z***, $\dot{z}$ and **u**.

The sampling time for the input voltage is T=**2** milliseconds (which is considered appropriate for obtaining an adequate discrete time model by applying the Euler approximation.)

Assume that the expected initial **z** and $\dot{z}$ are both equal to 0, and that the values of the initial **z** and $\dot{z}$ fall within the error of $\pm$ 0.01

You are required to provide the following equations:

1. Describing how to implement prediction steps, to be applied at that sampling rate **T.**
2. Describing how to implement EKF update steps when range, **r** and bearing, α measurements with reference to the LiDAR are available. Assume the case in which a landmark is detected, being that landmark at global position **(17,3)**. For this update step, assume the case in which, at the time of the EKF update, the position of the locomotive is z=**30** meters.

   It is assumed that the range measurement is polluted by WGN of std = **0.3** centimeters, and that the bearing measurement is affected by WGN of std= **0.4** degrees. It is also known that the LiDAR scans are generated every 160 milliseconds.
3. Now we have a speed sensor that monitors the speed of the vehicle. The speed sensor returns an analog signal, **b**, synchronous with the LiDAR scan. The relationship between the signal and speed is shown below.

$$\dot{z} = 0.4 * b^3 + 2 * b$$

3. Now we have a speed sensor that monitors the speed of the vehicle. The speed sensor returns an analog signal, ***b***, synchronous with the LiDAR scan. The relationship between the signal and speed is shown below.

$$\dot{z} = 0.4 * b^3 + 2 * b$$

The measurements of the signal **b** are polluted by WGN of standard deviation =**0.3**

You are required to propose a way to exploit these additional measurements, in the EKF process, for improving the estimates of the vehicle's state.

The marks allocation for this question is:

- 4 marks for question 1. (2 marks allocated on the correctness of the solution and 2 marks on the correct derivation and explanation.)
- 5 marks for question 2. (3 marks allocated on the correctness of the solution and 2 marks on the correct derivation and explanation.)
- 4 marks for question 3. (2 marks allocated on the correctness of the solution and 2 marks on the correct derivation and explanation.)

Your answer to this question must be provided in a file named "Q4.pdf" or "Q4.jpg". In the first lines of your file, you must write your name and ZID.

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